Economia das Organizações e dos Incentivos — MPE Insper

Exercise List: Real Book Exercises, Lecture 1

How to use this list. Every exercise below is a real exercise from one of the course books, reproduced verbatim: the source line gives author, book, chapter, exercise, the printed page and, in brackets, the page your PDF viewer shows for the file in `Livros/`, and whether the source gives the answer. Muñoz-Garcia & Toro-Gonzalez print a worked answer right after each exercise; MWG answers are in the MWG solutions manual in `Livros/`; Tadelis, Gibbons and Cabral give no answer in the book. When an exercise refers to a model, game or figure stated elsewhere in its book, that material is copied into the same block under "Referenced material", so every exercise can be solved where it is printed. There is one block per lecture of the real schedule. Each block has ten exercises in strictly rising difficulty, rated from 1 (apply a definition) to 5 (graduate level) next to each heading: three foundations, five exam level, two advanced. Try each exercise in the space provided before opening the book's answer.

Difficulty path. Lecture 1: from one 3×3 elimination (1) through two short proofs about dominance (Tadelis 4.1, MWG 8.B.4) to MWG 8.B.5 (5), where elimination over a continuum of quantities must be shown to converge with two firms and to stall with three.

Block	#	Source	p. [PDF]	Level	What it trains	Diff.	Answer
L1	1 (1)	Tadelis 4.5	77 [96]	Found.	IESDS round by round, 3×3 matrix	1	none
	2 (2)	Muñoz-Garcia Ch. 1 Ex. 13‡	26 [39]	Found.	Dominant and dominated strategies, then IDSDS	1	book
	3 (3)	Gibbons 1.2	48 [56]	Found.	IESDS, then pure Nash equilibria	2	none
	4 (4)	Tadelis 4.1‡	76 [95]	Exam	Proof: a strictly dominant-strategy equilibrium is unique	2	none
	5 (5)	Muñoz-Garcia Ch. 1 Ex. 12	23 [36]	Exam	Rationality vs. common knowledge of it	2	book
	6 (6)	Tadelis 4.7	77–78 [96–97]	Exam	Story to matrix, then elimination	3	none
	7 (7)	Tadelis 4.3‡	76 [95]	Exam	Discrete first-price auction: matrix, IESDS	3	none
	8 (8)	Tadelis 4.6	77 [96]	Exam	IESDS with continuous strategies	4	none
	9 (9)	MWG 8.B.4‡	262 [142]	Adv.	Proof: the order of strict deletion does not matter	4	manual
	10 (10)	MWG 8.B.5	262 [142]	Adv.	Cournot by IESDS: two firms vs. three	5	manual

#: item number in this lecture's list, then its number in the combined list (`lista-reais-lectures-01-05.pdf`) in brackets. Items not in the list for Lectures 1–4 (`lista-reais-lectures-01-04.tex`) are marked: † added in the 30-item version, ‡ added in the 50-item version. Every other item is reused from that list with its statement unchanged, some in a different block or position. "Diff.": difficulty from 1 (apply a definition) to 5 (graduate level). Pages: printed page, then the PDF viewer page in brackets (offsets in `Livros/book-registry.json`; MWG is a two-pages-per-sheet scan). "Answer": *book* = worked answer printed after the exercise; *manual* = MWG solutions manual in `Livros/`; *none* = no answer in the book. Books: Tadelis (2013), *Game Theory: An Introduction*; Gibbons (1992), *Game Theory for Applied Economists*; Muñoz-Garcia & Toro-Gonzalez (2019), *Strategy and Game Theory: Practice Exercises with Answers*, 2nd ed.; Mas-Colell, Whinston & Green (1995), *Microeconomic Theory* (MWG); Cabral (2017), *Introduction to Industrial Organization*, 2nd ed.

Block 1 — Static games I: dominance and iterated elimination *(Lecture 1)*

Exercise 1. (item 1 of the combined list) *Iterated Elimination* [Foundations; difficulty 1/5]

Source. Tadelis (2013), *Game Theory: An Introduction*, Ch. 4 "Rationality and Common Knowledge", Exercise 4.5 "Iterated Elimination", p. 77 [PDF 96]. No answer in the book. Reproduced verbatim.

4.5 Iterated Elimination: In the following normal-form game, which strategy profiles survive iterated elimination of strictly dominated strategies?

		Player 2		
		L	*C*	*R*
Player 1	*U*	6, 8	2, 6	8, 2
	M	8, 2	4, 4	9, 5
	D	8, 10	4, 6	6, 7

Resolution:

Exercise 2. (item 2 of the combined list) *Iterated Deletion of Strictly Dominated Strategies (IDSDS)* [Foundations; difficulty 1/5]

Source. Muñoz-Garcia & Toro-Gonzalez (2019), *Strategy and Game Theory: Practice Exercises with Answers*, 2nd ed., Ch. 1 "Dominance Solvable Games", Exercise 13 "Iterated Deletion of Strictly Dominated Strategies (IDSDS)"[A], p. 26 [PDF 39]. Worked answer in the book, right after the statement. Reproduced verbatim (statement only); Fig. 1.33, printed inside the answer on the same page, rebuilt.

Consider the following normal form game (Fig. 1.33)

(a) Find strictly dominant strategies (if any) for player 1 and for player 2.

(b) Find strictly dominated strategies (if any) for player 1 and for player 2.

(c) If you apply iterative deletion of strictly dominated strategies (IDSDS), what is the surviving strategy pair (or pairs)? Explain the steps you use in IDSDS, and why you use them.

		Player 2	
		Left	*Right*
Player 1	*Up*	-10, 10	0, 12
	Middle	-12, 0	2, 2
	Down	-9, 0	1, 0

Fig. 1.33 Normal-form game

Resolution:

Exercise 3. (item 3 of the combined list) *IESDS and pure-strategy Nash equilibria*

[Foundations; difficulty 2/5]

Source. Gibbons (1992), *Game Theory for Applied Economists*, Ch. 1 "Static Games of Complete Information", §1.5 Problems (Section 1.1), Problem 1.2, p. 48 [PDF 56]. No answer in the book. Reproduced verbatim; the payoff pair in cell (T, L) is printed "2.0" (comma typo) and is typeset "2, 0".

1.2. In the following normal-form game, what strategies survive iterated elimination of strictly dominated strategies? What are the pure-strategy Nash equilibria?

	L	C	R
T	2, 0	1, 1	4, 2
M	3, 4	1, 2	2, 3
B	1, 3	0, 2	3, 0

Resolution:

Exercise 4. (item 4 of the combined list) *Prove Proposition 4.1* [Exam level; difficulty 2/5]

Source. Tadelis (2013), *Game Theory: An Introduction*, Ch. 4 "Rationality and Common Knowledge", Exercise 4.1, p. 76 [PDF 95]. No answer in the book. Reproduced verbatim. The exercise states the proposition it asks to prove.

4.1 Prove Proposition 4.1: If the game $\Gamma = \langle N, \{S_i\}_{i=1}^n, \{v_i\}_{i=1}^n \rangle$ has a strictly dominant strategy equilibrium s^D, then s^D is the unique dominant strategy equilibrium.

Resolution:

Exercise 5. (item 5 of the combined list) *IDSDS and Rationality* [Exam level; difficulty 2/5]

Source. Muñoz-Garcia & Toro-Gonzalez (2019), *Strategy and Game Theory: Practice Exercises with Answers*, 2nd ed., Ch. 1 "Dominance Solvable Games", Exercise 12 "IDSDS and Rationality"[A], p. 23 [PDF 36]. Worked answer in the book, right after the statement. Reproduced verbatim (statement only).

Consider the following 3×3 matrix (Fig. 1.30):

		Player 2		
		F	C	B
Player 1	F	0, 5	2, 4	2, 3
	C	2, 3	0, 5	3, 2
	B	5, 0	2, 4	2, 3

Fig. 1.30 Normal-form representation of a three-players game

a. Does any player have a strictly dominant strategy?

b. Does any player have a strictly dominated strategy?

c. Use your results from part (b) to delete players' strictly dominated strategies assuming rationality, but not assuming common knowledge of rationality. [Hint: This means that every player can delete the strategies he finds to be strictly dominated but cannot put himself in the shoes of his opponent to delete the strategies his opponent finds to be strictly dominated.]

d. Use your results from part (b) to delete players' strictly dominated strategies assuming common knowledge of rationality. [Hint: This means that players can put themselves in the shoes of their opponents to delete strategies each other finds to be strictly dominated, as in the standard approach of IDSDS.]

Resolution:

Exercise 6. (item 6 of the combined list) *Campaigning* [Exam level; difficulty 3/5]

Source. Tadelis (2013), *Game Theory: An Introduction*, Ch. 4 "Rationality and Common Knowledge", Exercise 4.7 "Campaigning", pp. 77–78 [PDF 96–97]. No answer in the book. Reproduced verbatim.

4.7 Campaigning: Two candidates, 1 and 2, are running for office. Each has one of three choices in running his campaign: focus on the positive aspects of one's own platform (call this a positive campaign $[P]$), focus on the positive aspects of one's own platform while attacking one's opponent's campaign (call this a balanced campaign $[B]$), or finally focus only on attacking one's opponent (call this a negative campaign $[N]$). All a candidate cares about is the probability of winning, so assume that if a candidate expects to win with probability $\pi \in [0, 1]$, then his payoff is π. The probability that a candidate wins depends on his choice of campaign and his opponent's choice. The probabilities of winning are given as follows:

If both choose the same campaign then each wins with probability 0.5.

If candidate i uses a positive campaign while $j \neq i$ uses a balanced one then i loses for sure.

If candidate i uses a positive campaign while $j \neq i$ uses a negative one then i wins with probability 0.3.

If candidate i uses a negative campaign while $j \neq i$ uses a balanced one then i wins with probability 0.6.

a. Model this story as a normal-form game. (It suffices to be specific about the payoff function of one player and to explain how the other player's payoff function is different and why.)

b. Write down the game in matrix form.

c. What happens at each stage of elimination of strictly dominated strategies? Will this procedure lead to a clear prediction?

Resolution:

Exercise 7. (item 7 of the combined list) *Discrete First-Price Auction* [Exam level; difficulty 3/5]

Source. Tadelis (2013), *Game Theory: An Introduction*, Ch. 4 "Rationality and Common Knowledge", Exercise 4.3 "Discrete First-Price Auction", p. 76 [PDF 95]. No answer in the book. Reproduced verbatim.

4.3 Discrete First-Price Auction: An item is up for auction. Player 1 values the item at 3 while player 2 values the item at 5. Each player can bid either 0, 1, or 2. If player i bids more than player j then i wins the good and pays his bid, while the loser does not pay. If both players bid the same amount then a coin is tossed to determine who the winner is, and the winner gets the good and pays his bid while the loser pays nothing.

a. Write down the game in matrix form.

b. Does any player have a strictly dominated strategy?

c. Which strategies survive IESDS?

Resolution:

Exercise 8. (item 8 of the combined list) *Roommates* [Exam level; difficulty 4/5]

Source. Tadelis (2013), *Game Theory: An Introduction*, Ch. 4 "Rationality and Common Knowledge", Exercise 4.6 "Roommates", p. 77 [PDF 96]. No answer in the book. Reproduced verbatim.

4.6 Roommates: Two roommates each need to choose to clean their apartment, and each can choose an amount of time $t_i \geq 0$ to clean. If their choices are t_i and t_j, then player i's payoff is given by $(10 - t_j)t_i - t_i^2$. (This payoff function implies that the more one roommate cleans, the less valuable is cleaning for the other roommate.)

a. What is the best response correspondence of each player i?

b. Which choices survive one round of IESDS?

c. Which choices survive IESDS?

Resolution:

Exercise 9. (item 9 of the combined list) *Order of deletion* [Advanced; difficulty 4/5]

Source. Mas-Colell, Whinston & Green (1995), *Microeconomic Theory*, Ch. 8 "Simultaneous-Move Games", Exercises, Exercise 8.B.4^{C}, p. 262 [PDF 142, left half]. Answer in the MWG solutions manual (`Livros/`). Reproduced verbatim.

8.B.4^{C} Show that the order of deletion does not matter for the set of strategies surviving a process of iterated deletion of strictly dominated strategies.

Resolution:

Exercise 10. (item 10 of the combined list) *Cournot by successive elimination* [Advanced; difficulty 5/5]

Source. Mas-Colell, Whinston & Green (1995), *Microeconomic Theory*, Ch. 8 "Simultaneous-Move Games", Exercises, Exercise 8.B.5^{C}, p. 262 [PDF 142, left half]. Answer in the MWG solutions manual (`Livros/`; 8.B.5 is there). Reproduced verbatim. The pointer to Chapter 12 is context; the statement defines the whole model.

8.B.5^{C} Consider the Cournot duopoly model (discussed extensively in Chapter 12) in which two firms, 1 and 2, simultaneously choose the quantities they will sell on the market, q_1 and q_2. The price each receives for each unit given these quantities is $P(q_1, q_2) = a - b(q_1 + q_2)$. Their costs are c per unit sold.

(a) Argue that successive elimination of strictly dominated strategies yields a unique prediction in this game.

(b) Would this be true if there were three firms instead of two?

Resolution: